Exploring Two-Slit Interference - One Photon at a Time

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1 Introduction

Two-slit interference experiments have deep historical significance for our understanding of physics and the workings of the universe in general [1]. Isaac Newton first supported the corpuscular theory of light in 1704 with the publication of *Optiks* where he theorized that light was made of "corpuscles" that had some properties of particles. This view dominated until the 1800's when the two-slit experiment was first performed by Thomas Young. He seemingly disproved Newton's corpuscular theory of light when he discovered destructive interference properties of light that suggested that "light waves" existed. This light wave theory received even more support in the 19th century from dynamical theories of light such as those of Maxwell and Fresnel. Then, in the late 19th century, the photoelectric effect was observed. This effect was explained by Einstein using quanta of light, supporting the fact that light was made up of particles. This, along with other evidence for light particles known as "photons" such as blackbody radiation and the Compton effect threw into question Young's double slit results. Such questions gave rise to "wave-particle duality," or the thought that light behaves as both a wave and a particle. This experiment, according to Feynman, contains the heart of quantum mechanics and "the only mystery" [2].

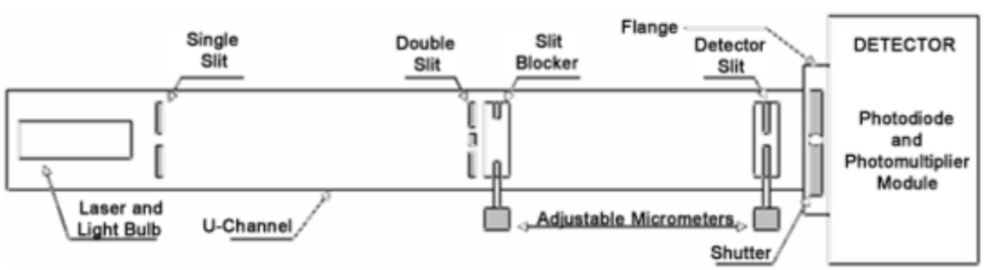

Figure 1: Diagram of the apparatus used in the experiment. Image credit: TeachSpin

Diving deeper into the mechanism behind this mystery underpins the experimental objectives for this experiment. We seek to demonstrate single photon interference, that is, the phenomenon of a single photon interfering with itself to create an interference pattern. We will also show that single-photon and multi-photon interference yield the same interference pattern. Finally, we will calculate the probability that one photon is present in the apparatus at a time. A diagram of the apparatus we will use to conduct the experiment is shown in 1.

2 Theory

When we perform the double slit experiment with light, we would expect one of two results[2]. We could find that light is quantized and made of particles, supported by results such as the photoelectric effect, and thus distributes like a particle would:

$$P_{12} = P_1 + P_2 \tag{1}$$

where P_{12} is the probability distribution from the center of the screen, and P_1 and P_2 are the distributions when the other hole is covered. We could also find that light behaves like a wave and interferes to create interference patterns described by:

$$I_1 = |h_1|^2 \quad I_2 = |h_2|^2 \quad I_{12} = |h_1 + h_2|^2, \tag{2}$$

where h_1 and h_2 are the amplitudes of the instantaneous height at the detector through hole 1 and 2 respectively.

When the experiment is finally run we find that the light does in fact interfere with itself and create interference patterns. For a single slit, we find the intensity as a function of anglular distance from the center of an idealized line source is:

$$I(\theta) = I(0)\left(\frac{sin(\left(\frac{\pi b}{2}sin(\theta)\right))}{\left(\frac{\pi b}{2}sin(\theta)\right)}\right)^2 \tag{3}$$

In our double slit scenario we use the Frauenhofer diffraction equation [3]

$$I\left(\theta\right) = 4I_0\left(\frac{sin^2\left(\frac{\pi b}{2}sin(\theta)\right)}{\left(\frac{\pi b}{2}sin(\theta)\right)^2}\right)cos^2\left(\frac{\pi a}{2}sin(\theta)\right), \tag{4}$$

where b is the slit width, a is the slit separation, and θ is the angular distance from the center, to describe the intensity of the light at the screen.

3 Methods

Setup and procedure from [1].

3.1 Materials

- Red laser
- Green filtered light bulb
- Photomultiplier tube (PMT) (used with light bulb)

- Photodiode (used with laser)
- Source slit
- Double slit
- Slit blocker
- Power source
- Small white paper screens
- Pulse counter
- Digital multimeter
- Oscilloscope
- Dim flashlight

3.2 Experimental Setup

The first part of our experimental setup is the alignment [4]. The first step is to remove all of the slits and set them flat side up and magnet side down. We then need to place the laser into a corner and adjust the base via the screw at the bottom until it is centered. With the laser in place, we then place the source slit so that it is at the center of the laser beam with the single slit diffraction pattern centerred on the detector aperture. Finally, we place the double slit so the single slit diffraction pattern from the source slit is centered on the double slit, and so that the double slit diffraction pattern is centered on the detector aperture. Next, we find and record five settings for the position of the slit blocker in which: 1) both slits are blocked, 2) the closer slit is blocked so light is only coming from the farther slit, 3) both slits are unblocked, 4) the farther slit is blocked so light is only coming from the closer slit, and 5) both slits are unblocked (this should correspond to the highest setting on the micrometer compared to the other settings). Then, we connect the digital multimeter. We then turn off our laser source and with the cover on, we record the reading on the multimeter. This is the zero offset that we will want to subtract from all of our readings output from the photodiode to ensure accuracy in our measurements.

After completing multi-photon interference measurements, we will want to set up the PMT and green light bulb for single-photon interference events. We do this by ensuring the bulb is aligned and using an oscilloscope to calibrate our PMT and pulse counter to count pulses that pass a certain threshold for measurements.

3.3 Experimental Procedure

3.3.1 Multi-photon Interference

With setup complete we are ready to take measurements. At first, we take coarse data before taking finer measurements near the maxima and minima. We will also want to take more measurements closer to the central maximum compared to the edges of the pattern. We do this with three different slit blocker placements: one slit open, the other slit open, and both slits open. Spacing of detector-slit positions can be determined by plotting the coarser data as we take it.

3.3.2 Single-photon Interference

For single-photon interference, we use the PMT to record the green filtered light bulb. The first set of data we take is to ensure that the PMT bias voltage is set to a suitable range. In this case our independent variable will be the PMT voltage and our dependent variable are both the count rates at the central maximum and when we close the shutter to the PMT. We then try both of these over range of 300 to 650 V of PMT bias and we plot these data on a semi-log plot. We should see the light plateau. With the PMT bias set, we record the photon count rate as a function of detector-slit position for each of one slit, the other slit, and both slits open. We also record this when both slits are blocked to record the background rate.

4 Safety Considerations

There are several safety considerations to take into account for this experiment. The first is laser safety. As we will be operating a laser to create multi-photon interference patterns, there are risks associated with the use of a laser. Our group will undergo the Laser Safety Training in BuckeyeLearn in order to ensure we are trained. We will take precautions by not pointing the laser outside of the plane of the experiment and not lowering our heads into the plane of the experiment, as well as removing all reflective apparel such as watches, jewelry, etc.

The other safety consideration is electrical safety. The PMT operates at a voltage of around 500-800 VDC. As such, we will take proper precautions to ensure no accidents occur.

5 Data Analysis and Interpretation

For the multi-photon interference data analysis, there will be three graphs of our data. There will be one of each of voltage-output signal versus detector-slit position for one slit open, the other slit open, and both slits open. We will then fit our data to equation 3 and 4 for single and double slit interference respectively.

For single-slit interference data analysis we will plot photon count rate versus detector-slit position for three cases: one slit open, the other slit open, and both slits open. We will then take our central maximum reading and the efficiency of the PMT to infer the arrival rate of all photons at the detector. We will use this arrival rate to calculate the average time interval between arrivals. Then, we compute the time it takes for a photon to travel the distance from the bulb to the detector, and compare this to the average time interval to compute the fraction of time there is one photon in the apparatus.

Sources of uncertainty include uncertainty in the position of the micrometers, uncertainty in the voltage of the PMT, uncertainty in the wavelength of the laser, and uncertainty in the slit width and separation gathered from fitting to the Frauenhofer diffraction equation.

6 Conclusions

In our conclusion we will conclude based on a statistical analysis that we have shown that only one photon at a time is present in the apparatus and interfered with itself. We will also provide the slit separation and width with uncertainties that we find from fitting.

References

[1] Two-Slit Interference, One Photon at a Time Operating Manual, Rev 3.0, TeachSpin, 2003.

[2] Feynman, R.P. (2015) The Feynman Lectures on Physics. Volume III, Quantum Mechanics. New York, New York: Basic Books.

[3] Hecht, E. (1998) Optics. Hecht. Reading, Mass: Addison-Wesley.

[4] 2-Slit Interference, 1 Photon at a Time v2.3, Carmen/Canvas, 2022.